

WEEK 3

HYPOTHESIS TESTING, EXPERIMENTS, DIFFERENCE IN MEANS

APPLIED STATISTICAL ANALYSIS/QUANTITATIVE METHODS I

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WHAT IS A HYPOTHESIS?

Definition

"A hypothesis is a statement about a population. It is a prediction that a parameter takes a particular numerical value or falls in a certain range of values" (Agresti and Finlay (2009, 143))

- We generally want to learn about population parameters using sample statistics
 - ▶ This process is known as statistical inference
- A central question is whether an observed sample result could plausibly arise from random sampling variation

COMPETING HYPOTHESES

Suppose we want to determine whether a population parameter differs from zero

H_0 : Population parameter equals zero

H_A : Population parameter differs from zero

- H_0 = null hypothesis
- H_A = alternative hypothesis
- Hypothesis tests assess evidence against H_0

THE FIVE STEPS OF HYPOTHESIS TESTING

TABLE 6.1: The Five Parts of a Statistical Significance Test

- | | |
|----|---|
| 1. | Assumptions
Type of data, randomization, population distribution, sample size condition |
| 2. | Hypotheses
Null hypothesis, H_0 (parameter value for “no effect”)
Alternative hypothesis, H_a (alternative parameter values) |
| 3. | Test statistic
Compares point estimate to H_0 parameter value |
| 4. | P-value
Weight of evidence against H_0 ; smaller P is stronger evidence |
| 5. | Conclusion
Report P -value
Formal decision (optional; see Section 6.4) |

General logic

Determine how unusual observed result would be if null hypothesis were true

Source: Agresti and Finlay (2009, 147)

STEP 1: ASSUMPTIONS

Important assumptions may concern:

- Type of data: continuous, categorical, or another form
- Sampling method: ideally a random sample
- Sample size: larger samples generally improve validity

Why assumptions matter

Validity of a test statistic and its reference distribution depends on assumptions being reasonably satisfied

STEP 2: STATE HYPOTHESES

Example

Someone claims that the mean income of a country is 1400

Suppose that:

- $n = 1000$
- $\bar{x} = 1350$
- $s = 750$

For a two-sided test:

$$H_0 : \mu = 1400$$

$$H_A : \mu \neq 1400$$

H_0 = population mean is equal to sample mean and H_A = the population mean is not equal to sample mean

ONE-SIDED AND TWO-SIDED HYPOTHESES

Hypotheses may be formulated as:

- One-sided hypotheses:

$$H_A : \mu > \mu_0 \quad \text{or} \quad H_A : \mu < \mu_0$$

- Two-sided hypotheses:

$$H_A : \mu \neq \mu_0$$

Interpretation

Form of H_A determines which parts of sampling distribution are considered extreme

STEP 3: TEST STATISTIC

Test statistic summarizes how far an estimate falls from value specified in null hypothesis

In many cases, this distance is expressed in standard-error units:

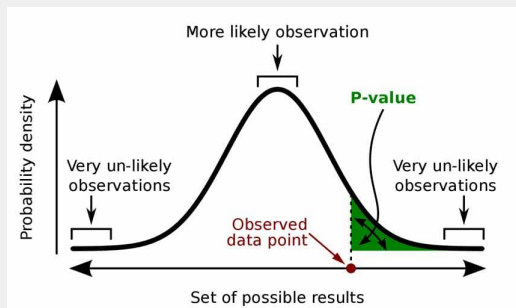
$$\text{Test statistic} = \frac{\text{estimate} - \text{null value}}{\text{standard error}}$$

- z-statistic is commonly used when population standard deviation is known or when large-sample approximations apply
- t-statistic is commonly used when sample is small or population variance is unknown

STEP 4: p -VALUE

Definition

p -value is probability of observing the realized test statistic, or a value even more extreme, assuming that H_0 is true



- Small p -value indicates that observed result is unusual under H_0
- Smaller p -value, stronger evidence against H_0
- p -value is not probability that H_0 is true!

STEP 5: CONCLUSION

Choose a significance level α , commonly:

$$\alpha = 0.05 \quad \text{or} \quad \alpha = 0.01$$

- If $p < \alpha$, reject H_0
- If $p \geq \alpha$, fail to reject H_0

Important

Rejecting H_0 does not prove that H_A is true! There remains a possibility of a false positive

TYPE I ERROR

- Type I error occurs when we reject null hypothesis even though it is true
- Ex: If $\alpha = 0.05$, we permit a 5% chance of rejecting a true null hypothesis under repeated sampling

		Null hypothesis (H_0) is	
		True	False
		Correct inference (true negative) (probability = $1-\alpha$)	Type II Error (false negative) (probability = β)
Decision about Null hypothesis (H_0)	Don't reject		
	Reject	Type I Error (false positive) (probability = α)	Correct inference (true positive) (probability = $1-\beta$)

CALCULATING A z-STATISTIC

Return to income example:

$$n = 1000, \quad \bar{x} = 1350, \quad \sigma = 750, \quad \mu_0 = 1400$$

Test statistic is:

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{1350 - 1400}{750 / \sqrt{1000}} \approx -2.11$$

$\alpha = 5\%$ for a two-sided test, critical values are $\approx \pm 1.96$

$$-2.11 < -1.96 \implies \text{reject } H_0$$

CONFIDENCE INTERVALS

A $100(1 - \alpha)\%$ confidence interval is an interval that would contain true population parameter in approximately $100(1 - \alpha)\%$ of repeated random samples

For a two-sided test with $\alpha = 0.05$, use a 95% confidence interval:

$$\bar{x} \pm z_{1-\alpha/2} \frac{s}{\sqrt{n}}$$

Connection to hypothesis tests

For a two-sided test at level α :

$$\mu_0 \notin \text{CI}_{100(1-\alpha)\%} \iff p < \alpha$$

EXAMPLE WITH A CONFIDENCE-INTERVAL

Using income example:

$$1350 \pm 1.96 \left(\frac{750}{\sqrt{1000}} \right) = [1303, 1396]$$

Since hypothesized mean, 1400, is not contained in 95% confidence interval, we reject H_0 at $\alpha = 0.05$

EX: POLITICAL IDEOLOGY

TABLE 6.2: Responses of Subjects on a Scale of Political Ideology

Response	Race		
	Black	White	Other
1. Extremely liberal	10	36	1
2. Liberal	21	109	13
3. Slightly liberal	22	124	13
4. Moderate, middle of road	74	421	27
5. Slightly conservative	21	179	9
6. Conservative	27	176	7
7. Extremely conservative	11	28	2
	$n = 186$	$n = 1073$	$n = 72$

$$H_0 : \mu = 4 \text{ (moderate)}$$

$$H_A : \mu \neq 4$$

H_A allows population mean to be either:

■ < 4 : more liberal population mean

■ > 4 : more conservative population mean

CALCULATING STANDARD ERROR

Suppose:

$$\bar{x} = 4.075, \quad s = 1.512, \quad n = 186$$

Standard error is:

$$SE = \frac{s}{\sqrt{n}} = \frac{1.512}{\sqrt{186}} \approx 0.111$$

Test statistic is:

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{4.075 - 4}{0.111} \approx 0.68$$

$$df = n - 1 = 185$$

How do we interpret this value?

INTERPRETING t -TEST

For a two-sided test with $\alpha = 0.05$ and $df = 185$:

$$t_{0.975,185} \approx 1.97$$

- Since $-1.97 < 0.68 < 1.97$ we fail to reject H_0
- p -value for two-sided test $p = \Pr(|T| > 0.68) \approx 0.50$
- 95% CI: $4.075 \pm 1.97(0.111) = [3.86, 4.29]$
- Therefore, we cannot reject H_0 and result is compatible with $\mu = 4$

DIFFERENCE-IN-MEANS t -TEST

A difference-in-means test evaluates whether the means of two groups differ.

The hypotheses are:

$$H_0 : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2$$

Equivalently:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_A : \mu_1 - \mu_2 \neq 0$$

DIFFERENCE-IN-MEANS TEST STATISTIC

Test statistic is:

$$t = \frac{(\bar{y}_1 - \bar{y}_2) - 0}{SE}$$

Under usual unequal-variance formulation:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

More generally:

$$t = \frac{\text{Estimate} - \text{Null value}}{\text{Standard error of estimate}}$$

EX: DIFFERENCE-IN-MEANS

TABLE 7.1: Cooking and Washing Up Minutes, per Day, for a National Survey of Men and Women Working Full Time in Great Britain

Sex	Sample Size	Cooking and Washing Up Minutes	
		Mean	Standard Deviation
Men	1219	23	32
Women	733	37	16

Observed group means:

$$\bar{y}_1 = 23, \quad \bar{y}_2 = 37$$

and standard error of their difference is:

$$SE = 1.09$$

Thus, test statistic is:

$$t = \frac{(37 - 23) - 0}{1.09} = \frac{14}{1.09} \approx 12.8$$

This is very far from zero!

INTERPRETING DIFFERENCE-IN-MEANS TEST

How likely would it be to observe a difference this large if two population means were equal?

- Resulting p -value is less than 0.05 and is essentially zero under reference distribution
- Since $p < 0.05$ we reject H_0 that two population means are identical
- Data provide strong evidence that population means differ
 - ▶ Sample means indicate that women report average household-work time $\neq$ men

Causal caution

Statistically significant difference between groups is not automatically a causal effect!

Ex: DO POLITICIANS RACIALLY DISCRIMINATE?

- Is racial discrimination a problem in the acquisition of government services?
- Do legislators discriminate against individual requests for constituency service on the basis of race?

DOING IT "RIGHT": EXPERIMENTAL DESIGN

Sample includes states legislators in 44 U.S. states with a valid e-mail address in September 2008¹

- Race was signaled by randomizing whether email was signed and sent from an email account with name Jake Mueller or DeShawn Jackson
- Text of email was also manipulated so as to signal partisan preference of the email sender
- Cross-tabulation between *race* & *partisan preference* gives six treatments (or groups)
- Outcome variable is response (or lack thereof) to any e-mails

¹Butler, D. M., & Broockman, D. E. (2011). Do politicians racially discriminate against constituents? A field experiment on state legislators. *American Journal of Political Science*, 55(3), 463-477.

EXAMPLE: DIFFERENCE OF MEANS

TABLE 1 Overall Effect Sizes—Does Jake Receive More Replies Than DeShawn?

	No Partisanship Signal	Republican Signal	Democratic Signal	Party Differential	
DeShawn	55.3%	54.3%	57.3%	-2.9%	Combined
Jackson	N = 806	N = 810	N = 812	($p = 0.23$)	-0.9%
Jake	60.5%	56.4%	55.3%	1.1%	($p = 0.61$)
Mueller	N = 812	N = 820	N = 799	($p = 0.31$)	
Race Differential	?	-2.1%	1.9%		
	($p = ?$)	($p = 0.39$)	($p = 0.43$)		

Let's calculate difference between the groups

EXAMPLE: CALCULATING DIFFERENCE OF MEANS

```
1 # create some fake data
2 deshawn <- rnorm(n=806, mean=55.3, sd=14.4)
3 jake <- rnorm(n=812, mean=60.5, sd=11.9)
```

EXAMPLE: CREATE OUR TS AND P-VALUE

```
1 # create function for "by hand"
2 # difference in means test
3 t.test.by.hand <- function(inputVec1, inputVec2, mo=0){
4   # get the means
5   m1 <- mean(inputVec1); m2 <- mean(inputVec2)
6   # standard deviations
7   s1 <- sd(inputVec1); s2 <- sd(inputVec2)
8   # number of observations
9   n1 <- length(inputVec1); n2 <- length(inputVec2)
10  # calculate standard error
11  se <- sqrt( (s1^2/n1) + (s2^2/n2) )
12  # calculate df using welch-satterthwaite df
13  df <- welchDF(s1, s2, n1, n2)
14  # since we have such a large sample, we'll get the same answer
15  # if we use pnorm instead
16  t <- (m1-m2-mo)/se
17  # create dataframe with all relevant info
18  dat <- c(m1-m2, se, t, 2*pt(-abs(t), df))
19  # make the table pretty
20  names(dat) <- c("Diff in means", "Std Error", "t", "p-value")
21  return(round(dat, 3))
22 }
```

EXAMPLE: INTERPRET RESULTS

```
1 # you'll find this output agrees with that of t.test when  
  you input x1,x2  
2 t.test.by.hand(deshawn, jake)
```

Diff in means	Std Error	t	p-value
-5.165	0.647	-7.982	0.000

```
1 # check to make sure our function works just as good  
2 t.test(deshawn, jake, var.equal = F)
```

t = -7.982, df = 1524.7, p-value = 2.809e-15
alternative hypothesis: true difference in means
is not equal to 0

CLASS BUSINESS

- Read required (and suggested) online materials
- Problem set # 1 is up on GitHub